ICPC Algorithm Library

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目录

1 Basic

1.1 test

```
export fs="-fsanitize=address,undefined"
ulimit -s 1024000
ulimit -v 1024000
mk() { g++ -o $1 $1.cpp -O2; }
```

1.2 pai

```
pai() {
  mk a; mk bf; mk gen
  while true; do
    ./gen > 1.in
    ./bf < 1.in > 1.ans
    ./a < 1.in > 1.out
    if diff -Zq 1.out 1.ans; then echo ac; else echo wa; break; fi
  done
}
```

1.3 run

```
shopt -s nullglob # 没有匹配到任何文件时，让它返回空
ulimit -s 1024000
g++ $1.cpp -o $1 -std=c++20 -O2 -fsanitize=address,undefined || exit
for f in "$2"/*.in; do
  x=${f%.in}
  \time -f "$x: %es %MKB" ./$1 < $x.in > $1.out
  diff -Zq $1.out $x.ans
done
```

2 Geometry

2.1 点和向量

```
using db = long double;
struct p2 {
  db x, y;
  db len() { return hypot(x, y); }
  p2 operator+(p2 b) { return {x + b.x, y + b.y}; }
  p2 operator-(p2 b) { return {x - b.x, y - b.y}; }
  p2 operator*(db k) { return {x * k, y * k}; }
  p2 operator/(db k) { return {x / k, y / k}; }
  db operator*(p2 b) { return x * b.x + y * b.y; } // 点积
  db operator%(p2 b) { return x * b.y - y * b.x; } // 叉积
};
using ps = vector<p2>;
```

2.2 线段和多边形

```
// 点到直线距离
db toL(p2 p, p2 a, p2 b) { return abs((b - a) % (p - a)) / (b - a).len(); }
```